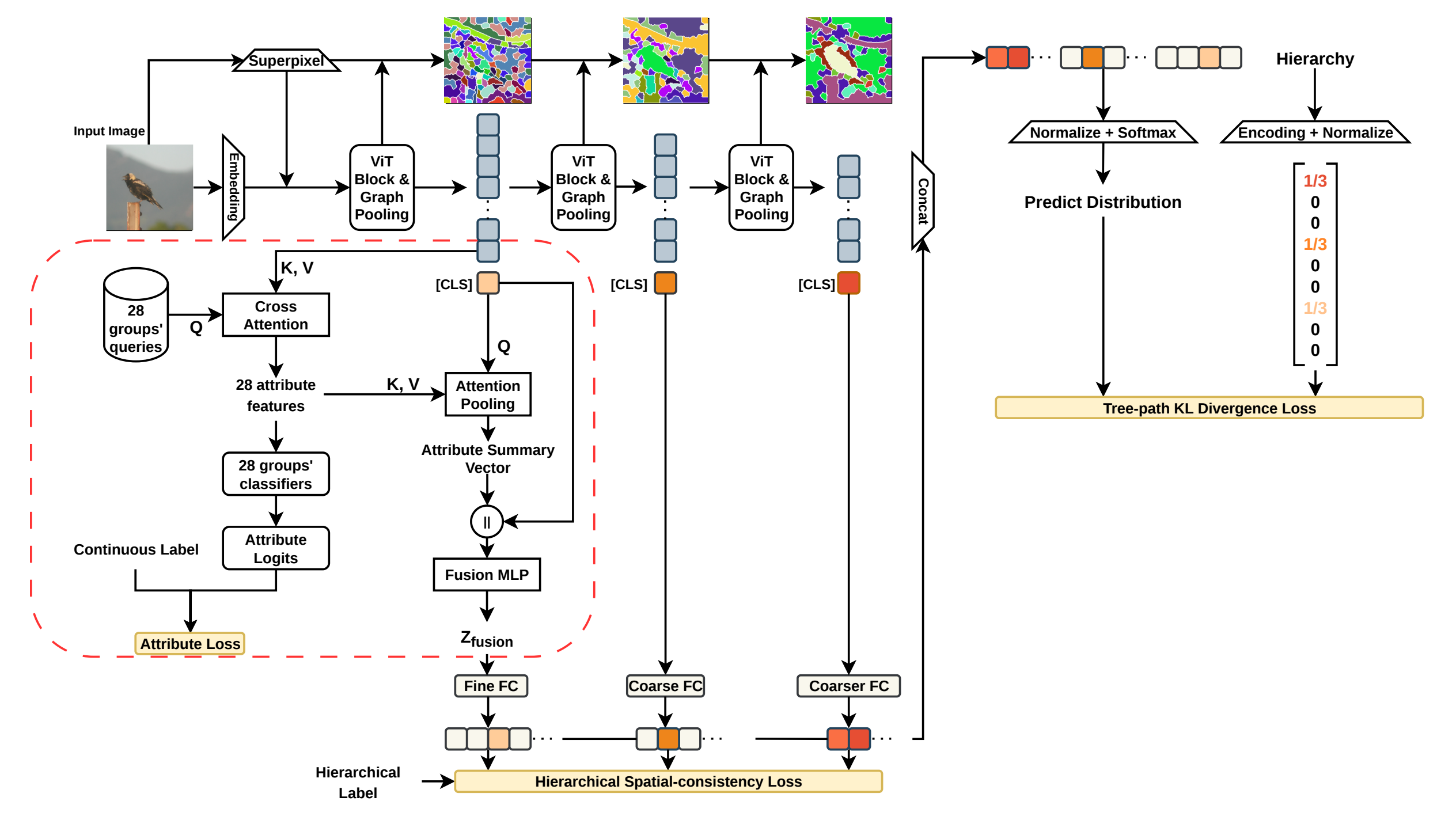

Input Image
Superpixel
Embedding
ViT Block & Graph Pooling
ViT Block & Graph Pooling
ViT Block & Graph Pooling
[CLS]
[CLS]
[CLS]
28 groups' queries
Q
Cross Attention
K, V
28 attribute features
K, V
Q
Attention Pooling
Attribute Summary Vector
28 groups' classifiers
Attribute Logits
Continuous Label
Attribute Loss
||
Fusion MLP
Z_{fusion}
Fine FC
Coarse FC
Coarser FC
Hierarchical Label
Hierarchical Spatial-consistency Loss
Concat
Normalize + Softmax
Predict Distribution
Hierarchy
Encoding + Normalize
1/3
0
0
1/3
0
0
1/3
0
0
Tree-path KL Divergence Loss